Naghmeh (Melody) Nazar

TRUSTWORTHY AI AGENTS | SPEECH MACHINE LEARNING | ROBUST EVALUATION

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RESEARCH PROFILE

Fall 2027 PhD applicant with an M.S. in Computer Engineering from California State University, Northridge. Current independent research examines indirect prompt injection in tool-using LLM agents and candidate pre-execution safeguards. Prior research developed a four-class speech-emotion model, with emphasis on transparent reporting of evaluation limits.

RESEARCH EXPERIENCE

Independent Research - Security of Tool-Using LLM Agents **2026 - present**

- Analyzing how untrusted email, document, and tool-output content can redirect an agent's proposed tool calls before execution.
- Conducting a structured analysis of the MCP-SafetyBench benchmark (Zong et al., ICLR 2026), mapping attack paths from untrusted context to tool selection and arguments.
- Evaluating confidence or uncertainty gating as a candidate baseline, not a validated defense; planned measures include attack success, benign completion, false-block rate, and calibration.

M.S. Thesis - Speech Emotion Recognition, CSUN **2024 - 2026**

- Developed a 2D ResNet-inspired CNN for neutral, happy, sad, and angry speech labels using RAVDESS and TESS.
- Processed speech at 16 kHz into 128 x 128 log-Mel spectrograms and trained with Adam, dropout, and residual connections.
- On the current split, achieved 80.36% validation accuracy and 74.4% on a held-out mixed test set.
- Speaker-independent and cross-corpus performance, desktop latency, and real-world generalization remain to be evaluated; no clinical validation is claimed.

EDUCATION

M.S. in Computer Engineering, California State University, Northridge **Aug 2024 - Aug 2026**

GPA: 3.44/4.0 | Thesis advisor: Prof. Shahnam Mirzaei

B.S. in Computer Engineering, University of Science and Technology of Mazandaran **Aug 2016 - Mar 2021**

RESEARCH OUTPUTS & PRESENTATION

- M.S. thesis research: Real-Time Speech Emotion Recognition Using Optimized Deep Residual Networks. Thesis PDF available on request.
- Presented the speech emotion recognition research at the 40th Annual CSU Student Research Competition. Materials available on request.

SELECTED ENGINEERING EVIDENCE

GenAI Email & Scheduling Agent **2026**

- Built an n8n tool-using email prototype and documented incomplete validation and approval boundaries around untrusted messages and generated actions.

Persian COVID-19 Social Behavior Analysis **2021 - 2023**

- Built sentiment, topic-modeling, and temporal-analysis pipelines to explore changes in Persian social-media posts.

PROFESSIONAL EXPERIENCE

Senior Frontend Developer, Golrang Industrial Group - Tehran, Iran **Jul 2022 - Jul 2024**

- Developed production React, React Native, Vue.js, and TypeScript applications; improved maintainability through profiling, caching, testing, and code review.

React Developer, Erole.ir - Tehran, Iran **Jun 2021 - Jun 2022**

- Built React/Redux applications and integrated REST APIs and SignalR-based real-time features.

METHODS & TECHNICAL FOUNDATION

Research & evaluation: experimental design, error analysis, threat modeling, reproducible evaluation, Git, Linux
Machine learning: Python, TensorFlow, Keras, scikit-learn, NumPy, Pandas, CNNs, ResNet
Signals & language: audio preprocessing, Mel-spectrograms, MFCCs, time-frequency analysis, NLP, topic modeling
Engineering: C++, JavaScript, TypeScript, React, Vue.js, Node.js, REST APIs